Rules of Inference

An **argument** is a sequence of statements, also called **premises**, that ends with a conclusion. Otherwise said, an argument contains one, or more, premises and a conclusion. A **valid argument** is an argument whose conclusion must be true because of the validity of the preceding statements. An argument is valid if, and only if, it is impossible for all the premises to be true and the conclusion to be false.

For example, suppose that an argument contains two statements: 1. If the light is red, then you must stop; and 2. The light is red. A valid argument, with these two premises, allows the conclusion – you must stop. This argument is valid because the truth of the two premises implies the truth of the conclusion.

Consider another argument that contains two statements: 1. If the light is red, then you must stop; and 2. The light is green. Suppose that in addition to these two premises, the conclusion – You can go-- is asserted. Is this a valid argument? Since it is possible to envision a circumstance for which both premises are true and the conclusion is false, this is not a valid argument.

The form of the first argument is:

$$p \to q$$
$$p$$
$$\therefore q$$

where p is the statement – The light is red, and q is the statement – You must stop. The first two lines are the premises of the argument, and third line is the consequence of the argument. The triangle of dots is read as, "therefore." The first premise is the statement – If the light is red, then you must stop. The second line is the statement – The light is red. Since both of these statements are presumed to be true, the logical consequence is that – You must stop.

This can also be seen with a truth table. Consider:

p	q	$p \to q$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**T**

Saying that $p \to q$ is true says that the first, third, or fourth row in the truth table is the case. Saying that p is true says that the first or second row in the truth table is the case. The only row for which both statements are true, for which the conjunction of these two statements ("adding" them together) is true, is the first row. What does this say about the truth value of q? It says that q is true. The

argument says that $(p \to q) \wedge (p)$ implies q. This argument, or rule of inference, is called modus ponens.

In addition to modus ponens, there are other rules of inference, including modus tollens, hypothetical syllogism, disjunctive syllogism, addition, simplification, conjunction, and resolution. The form of these rules of inference is always the same. Start with a set of premises that are assumed to be true. Given that the premises are true is sufficient to say that the conclusion is true.

Consider simplification, which is:

$$p \wedge q$$
$$\therefore p$$

If p and q is true, then you know that the truth values of both p and q are true. Specifically, p must be true. When you have many premises, multiple rules of inference may be required to create a valid argument that shows why the conclusion must be true.

There are additional rules of inference for quantified statements. For instance,

$$\forall x\, P(x)$$
$$\therefore P(c)$$

is called universal instantiation. Since $P(x)$ is true for all values x, it is true for a specific instance of $x = c$. Consequently, $P(c)$ must be true. Similarly, if $P(c)$ is true for an arbitrary value of c, then it must be true for all x. Consequently, $\forall x\, P(x)$ is true. This is called universal instantiation. There are similar rules of inference for the existential quantifier, existential instantiation, and existential generalization. The following table shows many useful rules of inference.

Rules of Inference

Rule of Inference	***Tautology***	***Name***
p $p \to q$ $\therefore q$	$(p \wedge (p \to q) \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism

$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \rightarrow (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \rightarrow p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

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Introduction to Proofs

A **proof** is a valid argument that shows the truth of some statement. Proofs use hypothesis or premises, axioms, and other proved theorems to show the truth of the resulting statement. **Axioms, postulates,** or **premises** are assumed to be true. A **theorem** is a statement that has a proof that shows it to be true.

Direct proofs start by assuming that a proposition p is true, and using rules of inference show that q must also be true. Direct proofs lead from the premise of a theorem to the conclusion for which the sequence of logical deductions is the steps in the proof.

Proof by contrapostion takes advantage of the logically equivalent statements.

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

(Construct a truth table to show the logical equivalence.) If you can prove the implication on the right had side of the equivalence is true, then you will have proved the implication on the left-hand side of the expression.

An implication, $p \rightarrow q$, is said to be **vacuously** true when the premise, p, is false. Recall that if p is false, then $p \rightarrow q$ is true regardless of the truth value of q. An implication, $p \rightarrow q$, is said to be **trivially** true if the conclusion, q, is already known to be true. Again recall that the cases in which q is true are those where $p \rightarrow q$ is true.

Proof by Contradiction is an indirect proof technique. The goal is to show that some statement p is true. To do so, look for a contradiction q such that $\neg p \rightarrow q$ is true. Because q is false and $\neg p \rightarrow q$ is true, this means that $\neg p$ is false, which means that p is true. Stated abstractly, this sounds more difficult to do than it actually is in practice. Practically speaking, you assume the negative of what is to be shown and then apply the rules of inference until an obvious contradiction is encountered. This argument is then the proof of what was desired to be proved.

Proof Methods and Strategy

Upon first encountering theorems and their proofs, each theorem and its proof may seem to be unique from other proofs of other theorems. There are, however, a few reoccurring strategies for proving theorems.

Exhaustive proofs are associated with theorems that can be shown to be true by examining a small number of cases. The cases taken together are said to exhaust all possibilities. In listing cases to be proved, you will often encounter symmetries that make the individual proofs of the symmetric cases essentially the same. You often invoke the statement **without loss of generality** then aggregate these cases together to be proved once.

Existence and **uniqueness** theorems and their proofs have special influence in mathematics. Many theorems assert that certain objects exist, such as a solution to an equation. These existence theorems take the form $\exists x P(x)$. You can show this statement to be true by producing some value x such that $P(x)$ is true. This type of proof is called a constructive proof. Nonconstructive existence proofs show that $\exists x P(x)$ is true by some other means. In particular, you might assume the converse and construct a proof by contraction.

Uniqueness proofs add to existence proof the requirement that there only is one object that satisfies the assertion. These types of proofs begin by proving the existence of an object, x. Then you show that if $y \neq x$, and then y do not have the desired property. Alternatively, if you can, show that both x and y have the desired property that $x = y$.